

Chapter 4: Random Variables (Sections 4.6 – 4.7)

4.6 The Bernoulli and Binomial Random Variables

Bernoulli Random Variable

A random variable X representing a single trial with two outcomes (Success/Failure).

- $X = 1$ if success (prob p), $X = 0$ if failure (prob $1 - p$).
- **PMF:**

$$p(0) = 1 - p, \quad p(1) = p$$

Binomial Random Variable

Represents the number of successes in n independent Bernoulli trials, each with probability p .

- **Parameters:** (n, p)
- **PMF:**

$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}, \quad i = 0, 1, \dots, n$$

- **Validity:** The sum of probabilities equals 1 by the Binomial Theorem: $\sum_{i=0}^n p(i) = [p + (1 - p)]^n = 1$.

Key Properties

- **Expected Value:** $E[X] = np$
- **Variance:** $\text{Var}(X) = np(1 - p)$
- **Shape:** The PMF increases monotonically and then decreases.
 - The mode (value with highest probability) occurs at the largest integer $\leq (n + 1)p$.
 - If $(n + 1)p$ is an integer, there are two modes: $(n + 1)p$ and $(n + 1)p - 1$.

Recursive Formula

Useful for computing probabilities efficiently:

$$P(X = k + 1) = \frac{p}{1 - p} \cdot \frac{n - k}{k + 1} \cdot P(X = k)$$

Examples

1. **Coin Flips:** 5 flips, prob of heads $1/2$. $P(X = 0) = 1/32$, $P(X = 1) = 5/32$, etc.
2. **Defective Items:** In a package of 10 items with defect rate 0.01, prob of replacing package (if > 1 defect) is $1 - P(0) - P(1) \approx 0.004$.
3. **Wheel of Fortune:** Betting on a number with 3 dice ($n = 3, p = 1/6$). Expected return is negative ($-17/216$), meaning the player loses on average.
4. **Voting Power:** In a state of size n , the "power" of a voter (probability of breaking a tie) is proportional to $1/\sqrt{n}$. Thus, expected electoral influence is proportional to $\sqrt{n}$ (favoring larger states).

4.7 The Poisson Random Variable

Definition

A random variable X taking values $0, 1, 2, \dots$ is Poisson with parameter $\lambda > 0$ if:

$$P(X = i) = e^{-\lambda} \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

Key Properties

- **Expected Value:** $E[X] = \lambda$
- **Variance:** $\text{Var}(X) = \lambda$
- **Recursive Formula:**

$$P(X = i + 1) = \frac{\lambda}{i + 1} P(X = i)$$

Poisson Approximation to Binomial

The Poisson distribution approximates the Binomial distribution when n is large and p is small, such that $\lambda = np$ is moderate.

- **Approximation:**

$$\binom{n}{i} p^i (1-p)^{n-i} \approx e^{-\lambda} \frac{\lambda^i}{i!}$$

- **Applications:** Misprints in a book, rare diseases, alpha particle emissions, surviving to age 100.
- **Poisson Paradigm:** Applies even when trials are "weakly dependent" (e.g., matching problem, birthday problem).

Poisson Process

Models events occurring continuously in time (e.g., earthquakes, customers arriving) if:

1. Prob of 1 event in small interval h is $\approx \lambda h$.
 2. Prob of ≥ 2 events in h is negligible.
 3. Events in non-overlapping intervals are independent.
- **Result:** Number of events in time t , $N(t)$, is Poisson with mean λt .

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$$

- **Waiting Time:** The time until the next event, X , has an exponential distribution: $P(X > t) = e^{-\lambda t}$.

Examples

1. **Typos:** If typos per page are Poisson($\lambda = 0.5$), prob of at least one error is $1 - e^{-0.5} \approx 0.393$.
2. **Radioactive Decay:** Alpha particles in 1 second ($\lambda = 3.2$). Prob of ≤ 2 particles is ≈ 0.3799 .
3. **Longest Run:** The length of the longest run of heads in n coin flips can be analyzed using a Poisson approximation for the number of runs of length k .
4. **Birthday Problem (Approximation):** Number of pairs with same birthday is approx Poisson with $\lambda = \binom{n}{2}/365$. For $P(\text{no match}) < 0.5$, we need $e^{-\lambda} < 0.5 \Rightarrow n \approx 23$.